



Syllabus for GOV/STAT 618: Bayesian Statistics

American University

Term: 2026 Fall Semester

Wednesday 5:30 PM - 8:00 PM

Location: Kerwin Hall Classroom 107

Instructor

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Thursdays from 2-4 on Zoom

Description. This course introduces the theoretical and applied foundations of Bayesian statistical analysis focused on the problems encountered by scientists. The Bayesian paradigm is ideally suited to the type of data analysis required of social and biomedical scientists because it recognizes the mobility of population parameters, incorporates prior knowledge that researchers possess, and updates estimates as new data are observed. This course will introduce the basic principles of Bayesian statistics to students in the social and behavioral sciences, focusing on examples drawn from sociology, political science, economics, marketing, psychology, public policy, and anthropology. Models will be estimated with bugs/jags as well as basic R. Therefore course will make exclusive use of software that is in the public domain, yet is high in quality. Most of the examples will be drawn from sociology, political science, economics, marketing, psychology, public policy, and anthropology. The prerequisites for this course are a linear regression course, exposure to calculus, and knowledge of basic matrix algebra. In this course you will be expected to: complete 6 problem sets, submit on Canvas, participate in the course via lectures, sections, and discussion forums, complete a final group research project, and provide feedback on another group's research project.

Course Objectives. After taking this course you will:

- Understand the basic issues in Bayesian inference that are applied to all Bayesian models.

- Be able to critically evaluate the assumptions that justify modeling decisions and techniques in Bayesian statistical specifications.
- Have the ability to implement Bayesian regression models with applied data and dedicated software from R.

Prerequisites. Students in the class should plan to work hard at understanding some difficult material. You'll need to have proficiency with R. You should have taken DATA 614 or the equivalent. Of course there are always exceptions to this, so feel free to talk with me about your background to see if the fit is right.

Grading.

- Biweekly homework assignments (60% of final grade)
- Final project (30% of final grade)
- Participation (10% of final grade)

Course Structure. Lectures will cover the basic theoretical issues in class along with some applied examples. We'll try to foster discussions when we can and perhaps even attempt some "breakout" time to work through issues. Your attendance at lectures is very strongly recommended and your participation counts toward your grade.

Reading. There are readings for each topic and they mostly cover the theory of the method along with some applications. Obviously, read the required readings and any others that pique your curiosity. In addition, though, engage with the readings: take notes, write down your impressions or confusions, talk with your classmates. All of your classes should be pushing your research forward and you will be more creative the more you actively read. All slides are available on the **Canvas** site for the course and at <https://github.com/jgill122/GOV.STAT-2017->. Some data will be available at the Github page as well. Every dataset in the book is in the R package **BaM**.

Course Assignments. Participation is crucial component to the class and we will grade your participation based on your engagement during class meetings along with your contributions to discussions and feedback via **Canvas**. Methods are tools and it is not very instructive to read a lot about hammers or watch someone else wield a hammer. You need to get your hands on a hammer or two. Thus, in this course, you will have 6 problem sets. They will be

a mix of conceptual questions, analytic problems, computer simulations, and data analysis. These problem sets should be typed and well-formatted, using **RMarkdown**. Each problem set will be equally weighted. **NO LATE ASSIGNMENTS WILL BE ACCEPTED**. I encourage you to work in groups on the homework, but you always need to write your own solutions including your computer code. This is something we routinely check. Also, it is hugely beneficial to attempt the problems sets on your own before working in groups. The schedule for the problem sets is below in the calendar.

Student Project. In lieu of a final exam, this course requires students to write a short paper applying or extending the Bayesian methods we learn in this class. These papers can be substantive (applying the methods to a problem of interest to you) or methodological (deriving a new Bayesian method that innovates over existing ones). It should be no longer than 20 double-spaced pages and focus on the research design, data, methodology, results, and analysis. Literature reviews or background material should be omitted or included in an appendix. This is a group project. More details to be given in class.

Books. We will use the following text for this class:

- Jeff Gill, *Bayesian Methods for the Social and Behavioral Sciences*, THIRD EDITION. Chapman & Hall/CRC. 2014.

Course Plan

- Week 1, September 2, Background and Introduction:
 - Problems with non-Bayesian social science statistics
 - General Motivation and Justification
 - Bayes' Law
 - Conditional Inference with Bayes' Law
 - Statistical Models with Bayes' Law
 - Bayesian Statistics as a Scientific Approach to Social and Behavioral Data Analysis
 - Introducing Markov Chain Monte Carlo Techniques
 - Simple Gibbs Sampling
 - Bayes Factor

Problem Set # 1. Due September 16: 1.3, 1.8, 1.10, 1.12, 2.1, 2.3, 2.5, 2.16.

- Week 2, September 9, Linear Model Theory

- Week 3, September 16, Specifying Bayesian Models:

- Likelihood Theory and Estimation
- The Basic Bayesian Framework
- Developing the Bayesian Inference Engine
- Summarizing Posterior Distributions with Intervals
- Bayesian Credible Intervals
- Bayesian Highest Posterior Density Intervals
- Quantile Posterior Summaries
- Beta-Binomial Model
- Bayesian “Learning”

Problem Set # 2. Due September 30: 3.6, 3.7, 3.11, 3.19, 4.5, 4.7, 4.15, 4.20.

- Week 4, September 23, The Normal and Student’s-t Models:

- The Normal Model with Variance Known
- The Normal Model with Mean Known
- The Normal Model with Both Mean and Variance Unknown
- Multivariate Normal Model, Mean and Variance Both Unknown
- Simulated Effects of Differing Priors
- The Student’s-t Model
- Normal Mixture Models

- Week 5, September 30, The Bayesian Prior:

- Conjugate Prior Forms
- The Exponential Family Form
- Limitations of Conjugacy
- Uninformative Prior Distributions
- Uniform Priors
- Jeffreys Prior
- Reference Priors
- Improper Priors
- Informative Prior Distributions
- Power Priors
- Elicited Priors
- Hybrid Prior Forms
- Spike and Slab Priors for Linear Models
- Maximum Entropy Priors
- Histogram Priors
- Nonparametric Priors
- Bayesian Shrinkage

Problem Set # 3. Due October 14: 5.5, 5.9, 5.12, 5.19, 6.4, 6.9, 6.11, 6.13

- Week 6, October 7, The Bayesian Linear Model:

- The Basic Regression Setup
- Uninformative Priors for the Linear Model
- Conjugate Priors for the Linear Model
- Conjugate Caveats for the Cautious and Careful
- Posterior Predictive Distribution for the Data
- Linear Regression with Heteroscedasticity

Note: Professor Ryan Moore will be covering the key principles of Causal Inference in class.

- Week 7, October 14, The Basics of Markov Chain Monte Carlo:

- What Is a Markov Chain?
- A Markov Chain Illustration
- The Chapman-Kolmogorov Equations
- Marginal Distributions
- General Properties of Markov Chains
- Homogeneity
- Irreducibility
- Recurrence
- Stationarity
- Ergodicity
- The Gibbs Sampler
- Description of the Algorithm
- Handling Missing Dichotomous Data with the Gibbs Sampler
- Summary of Properties of the Gibbs Sampler

Problem Set # 4. Due October 28: 7.5, 7.7, 7.9, 7.19, 10.7, 10.9, 10.13

- Week 8, October 21, More Samplers:

- The Metropolis-Hastings Algorithm
- Background
- Description of the Algorithm
- Metropolis-Hastings Properties
- Metropolis-Hastings Derivation
- The Transition Kernel
- The Hit-and-Run Algorithm

- Week 9, October 28, Implementing Bayesian Models with Markov Chain Monte Carlo, Part 1:

- Basic BUGS Syntax

- Constructing a model in **JAGS**
- Running **JAGS** from **R**
- The Deviance Information Criterion

Problem Set # 5. Due November 11: 10.11, 10.15, 10.17, 12.3, 12.9, 12.17, 12.19

- Week 10, November 4, Implementing Bayesian Models with Markov Chain Monte Carlo, Part 2:
 - Assessing Convergence of Markov Chains
 - Graphical Diagnostics
 - Standard Empirical Diagnostics
 - The Geweke Time-Series Diagnostic
 - Gelman and Rubin's Multiple Sequence Diagnostic
 - The Heidelberger and Welch Diagnostic
 - Summary of Diagnostic Similarities and Differences
- Week 11, November 11, Bayesian Hierarchical Models:
 - Basic Multilevel Models
 - Comparing Variances
 - Structure of the Bayesian Version
 - Priors and Hyperpriors in a Hierarchical Setting
 - Bayesian Nonlinear Regression Specifications

Problem Set # 6. Due November 18: 14.1, 14.5, 14.7, 14.11

- Week 13, November 18, Nonparametric Bayesian Modeling:
 - Some required measure theory
 - Dirichlet processes for Bayesian Models
 - The mathematics of clustering
 - A worked example

Work on Projects

- Week 14, November 25, No Class.
- Week 15, December 2, Classroom Presentation of Models:
 - Model summaries due on **Canvas**.